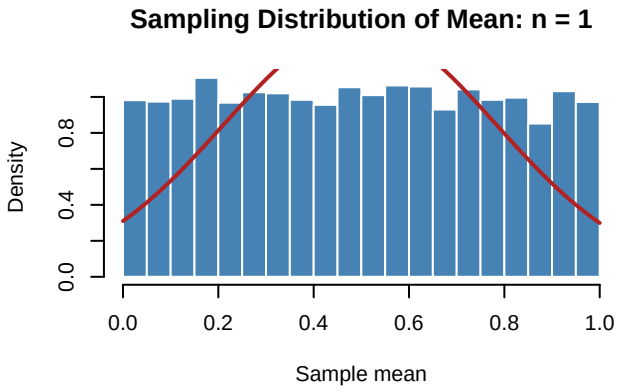
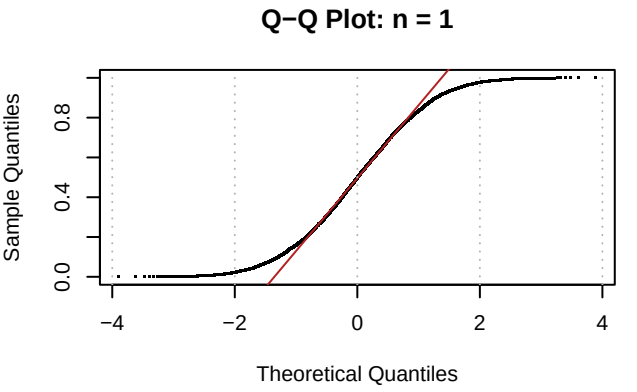
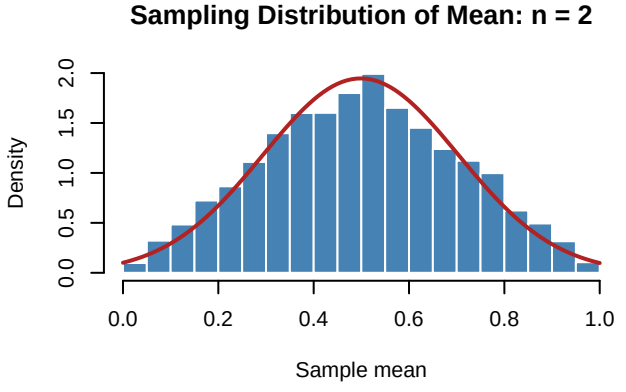
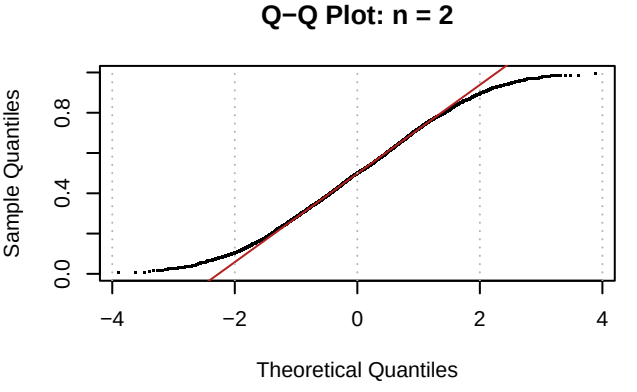
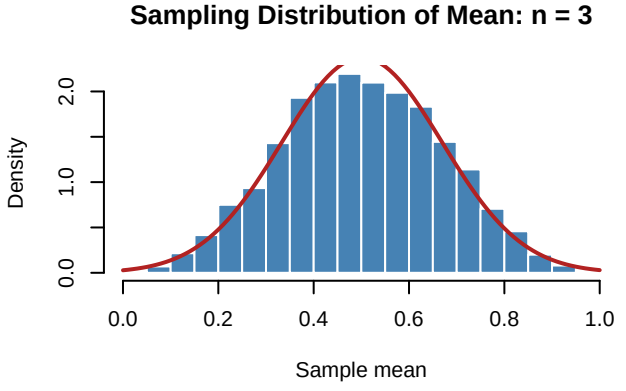
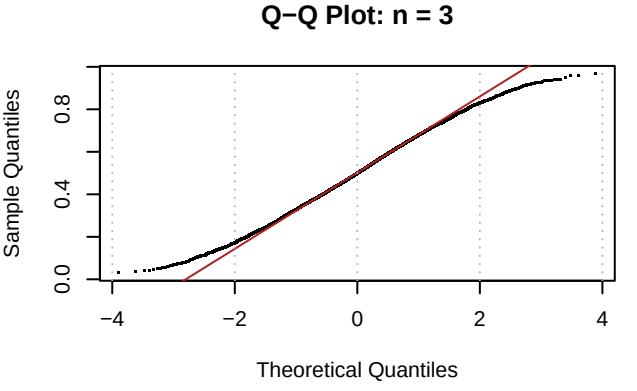

Sampling Distribution of Mean: n = 1
Density
0.0
0.4
0.8
0.0
0.2
0.4
0.6
0.8
1.0
Sample mean

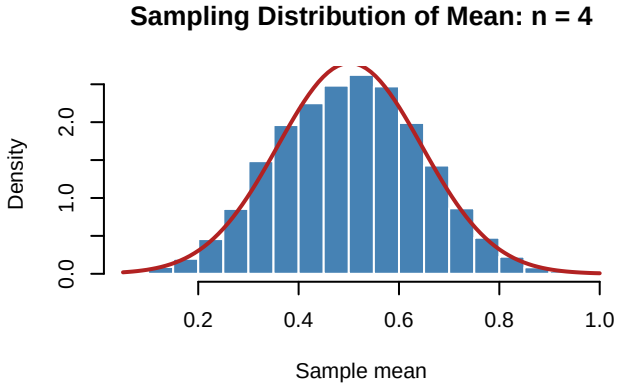

Q−Q Plot: n = 1
Sample Quantiles
0.0
0.4
0.8
−4
−2
0
2
4
Theoretical Quantiles

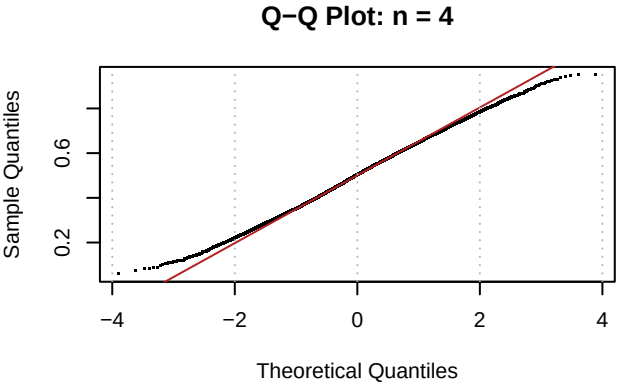

Sampling Distribution of Mean: n = 2
Density
0.0
0.5
1.0
1.5
2.0
0.0
0.2
0.4
0.6
0.8
1.0
Sample mean

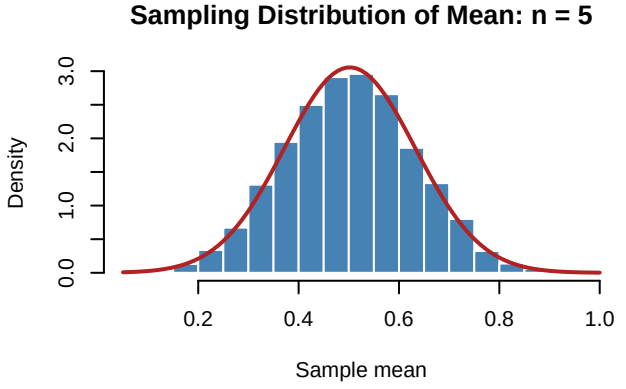

Q−Q Plot: n = 2
Sample Quantiles
0.0
0.4
0.8
−4
−2
0
2
4
Theoretical Quantiles

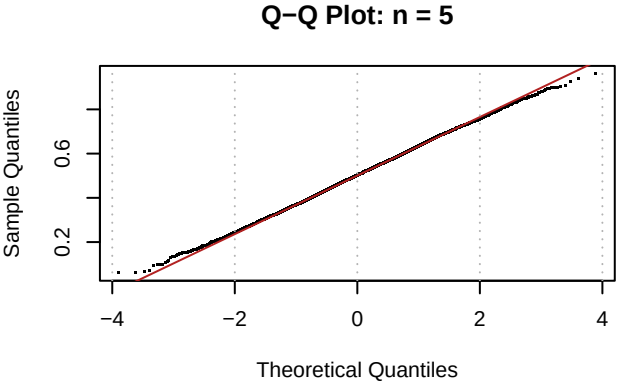

Sampling Distribution of Mean: n = 3
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
Sample mean

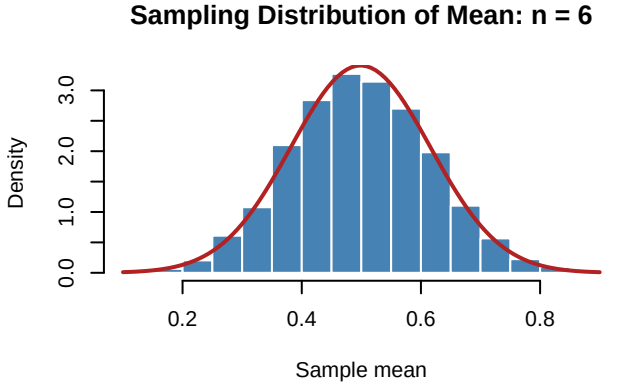

Q−Q Plot: n = 3
Sample Quantiles
0.0
0.4
0.8
−4
−2
0
2
4
Theoretical Quantiles

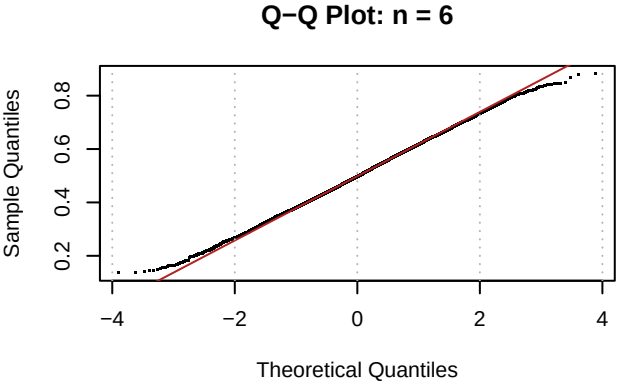

Sampling Distribution of Mean: n = 4
Density
0.0
1.0
2.0
0.2
0.4
0.6
0.8
1.0
Sample mean

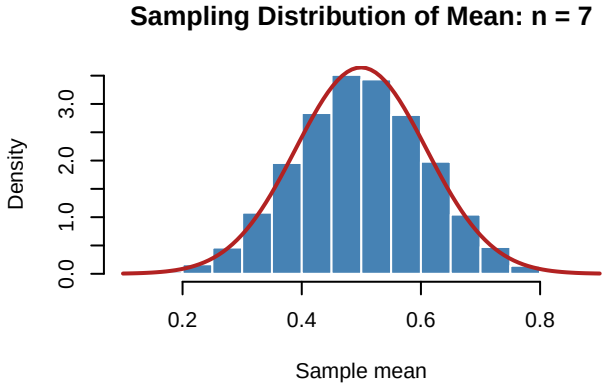

Q−Q Plot: n = 4
Sample Quantiles
0.2
0.6
−4
−2
0
2
4
Theoretical Quantiles

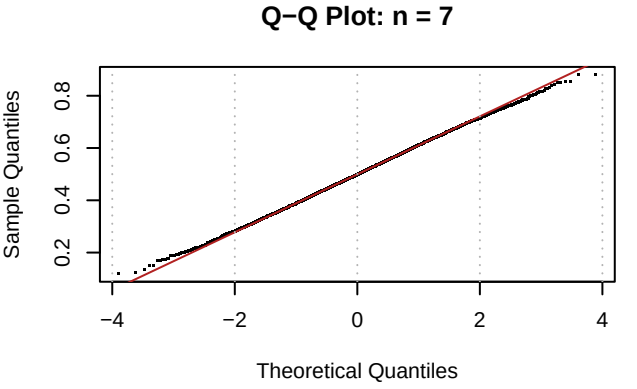

Sampling Distribution of Mean: n = 5
Density
0.0
1.0
2.0
3.0
0.2
0.4
0.6
0.8
1.0
Sample mean

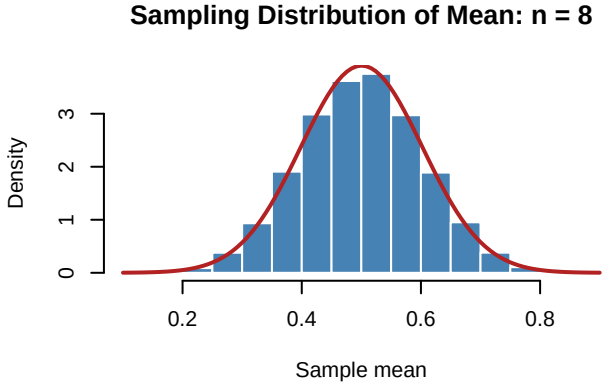

Q−Q Plot: n = 5
Sample Quantiles
0.2
0.6
−4
−2
0
2
4
Theoretical Quantiles

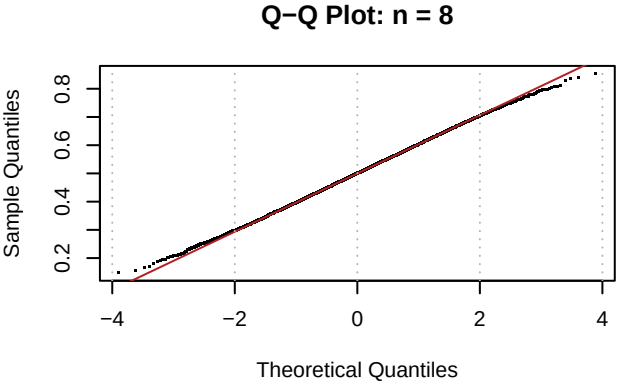

Sampling Distribution of Mean: n = 6
Density
0.0
1.0
2.0
3.0
0.2
0.4
0.6
0.8
Sample mean

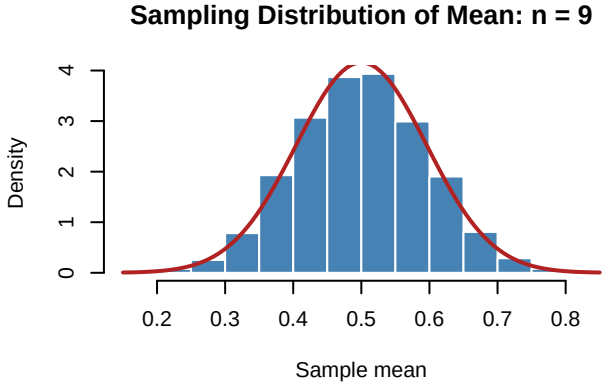

Q−Q Plot: n = 6
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

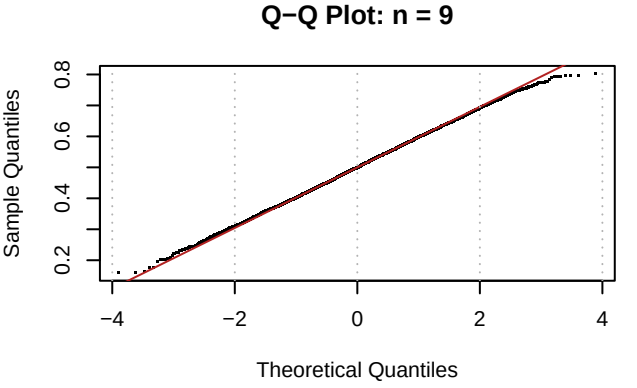

Sampling Distribution of Mean: n = 7
Density
0.0
1.0
2.0
3.0
0.2
0.4
0.6
0.8
Sample mean

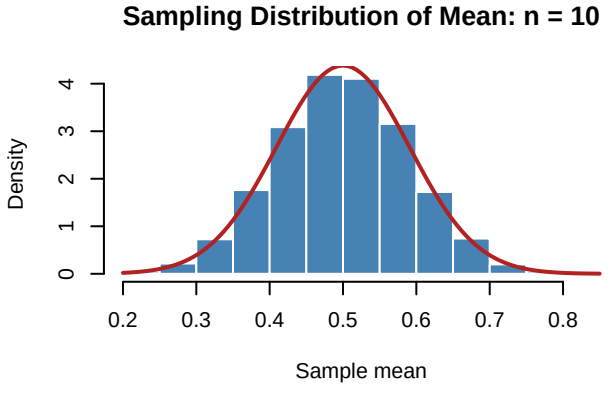

Q−Q Plot: n = 7
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

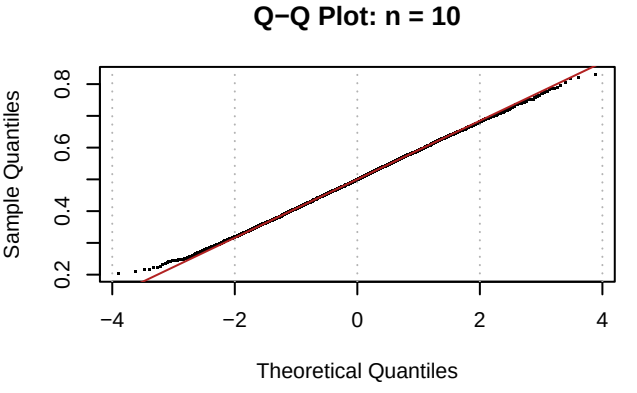

Sampling Distribution of Mean: n = 8
Density
0
1
2
3
0.2
0.4
0.6
0.8
Sample mean

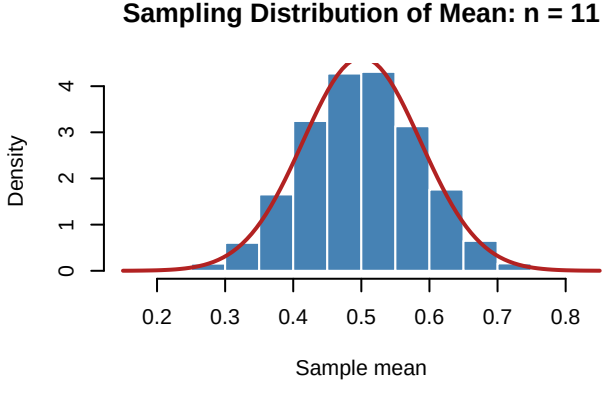

Q−Q Plot: n = 8
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

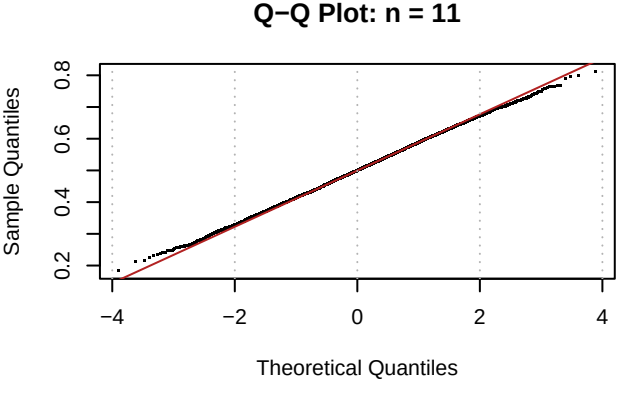

Sampling Distribution of Mean: n = 9
Density
0
1
2
3
4
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

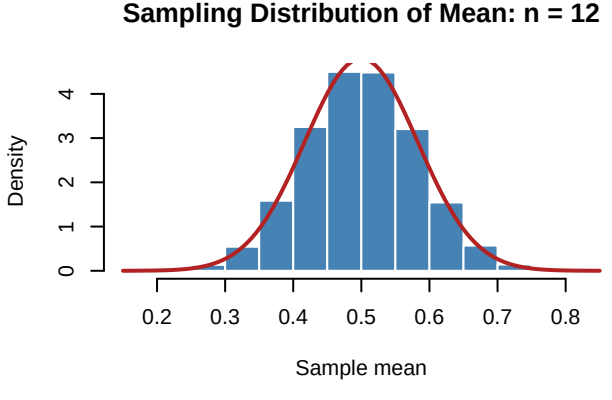

Q−Q Plot: n = 9
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

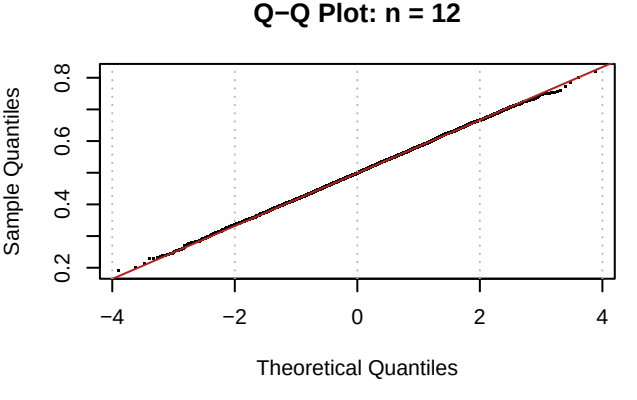

Sampling Distribution of Mean: n = 10
Density
0
1
2
3
4
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

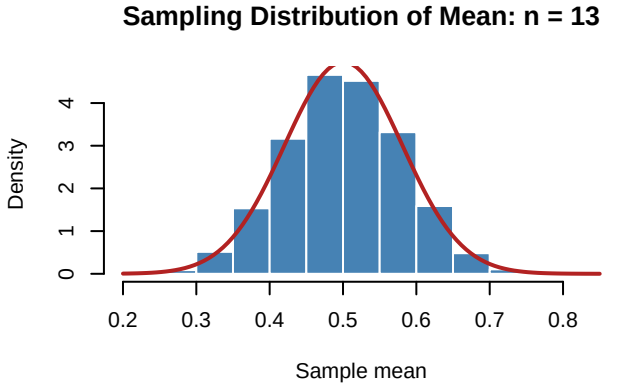

Q−Q Plot: n = 10
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

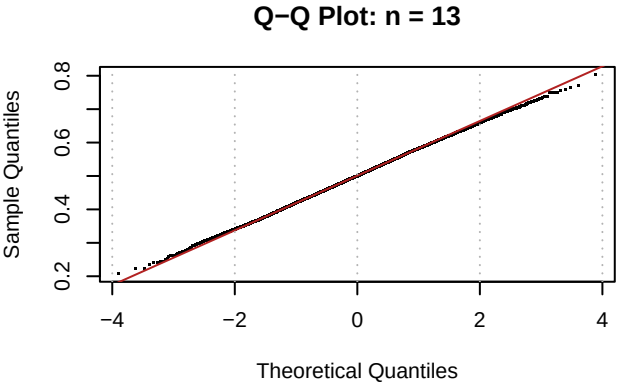

Sampling Distribution of Mean: n = 11
Density
0
1
2
3
4
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

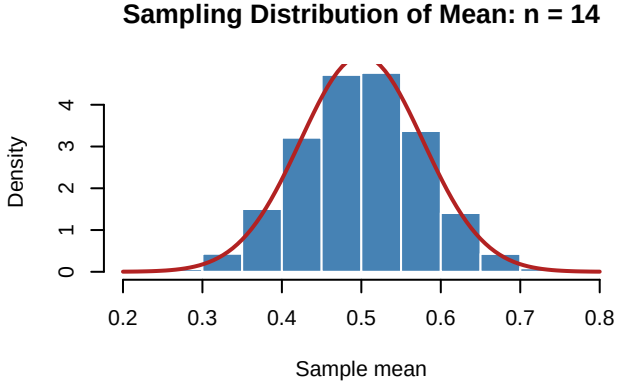

Q−Q Plot: n = 11
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

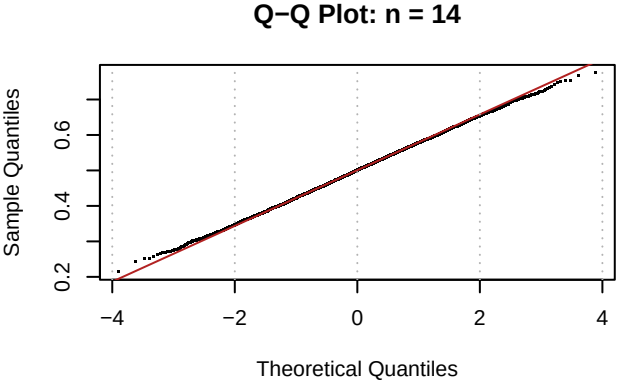

Sampling Distribution of Mean: n = 12
Density
0
1
2
3
4
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

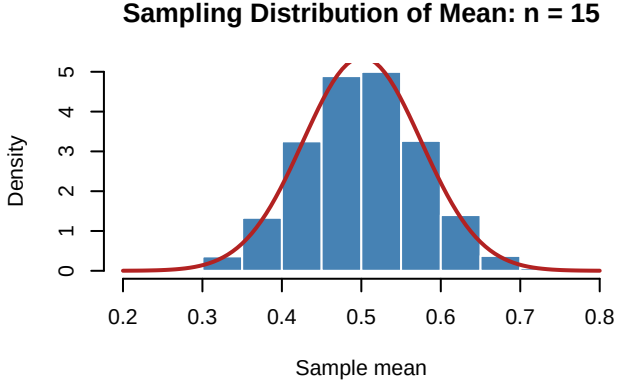

Q−Q Plot: n = 12
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

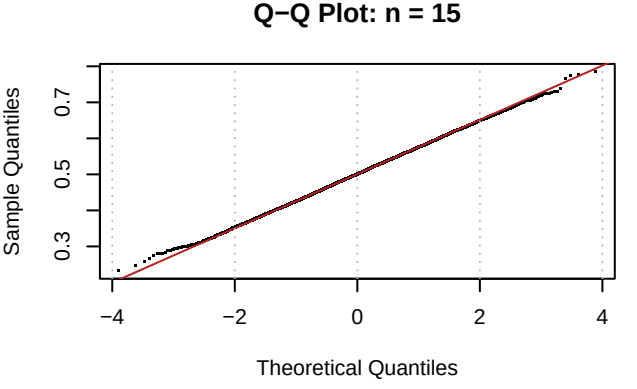

Sampling Distribution of Mean: n = 13
Density
0
1
2
3
4
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

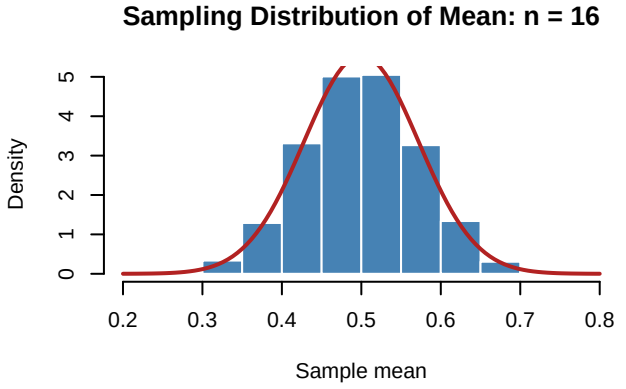

Q−Q Plot: n = 13
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

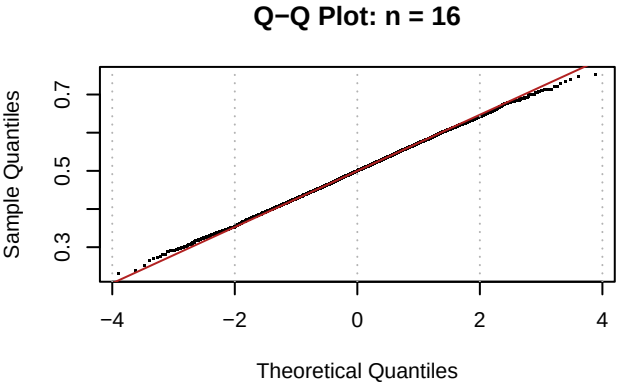

Sampling Distribution of Mean: n = 14
Density
0
1
2
3
4
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

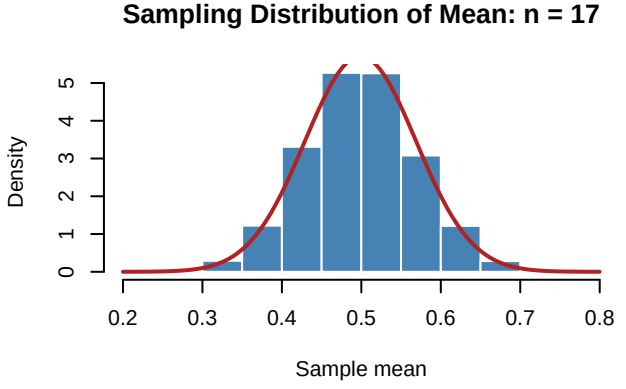

Q−Q Plot: n = 14
Sample Quantiles
0.2
0.4
0.6
−4
−2
0
2
4
Theoretical Quantiles

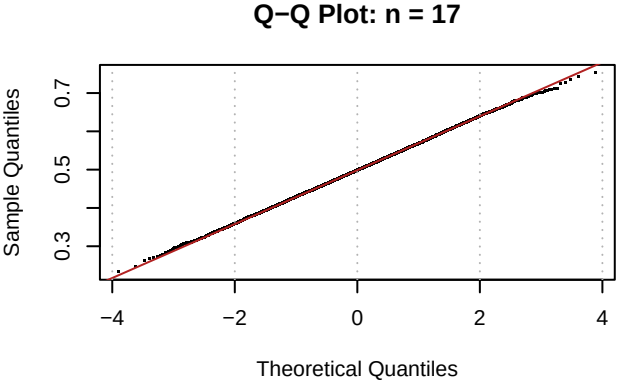

Sampling Distribution of Mean: n = 15
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

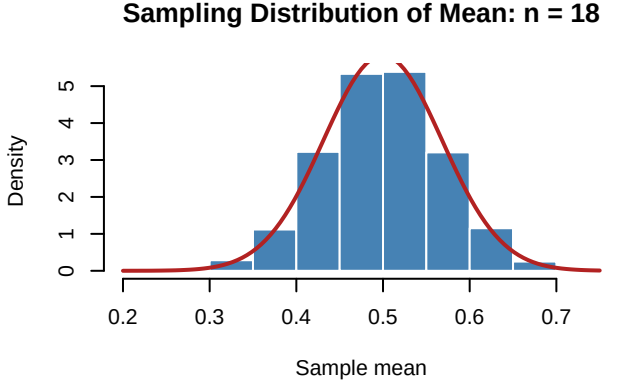

Q−Q Plot: n = 15
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

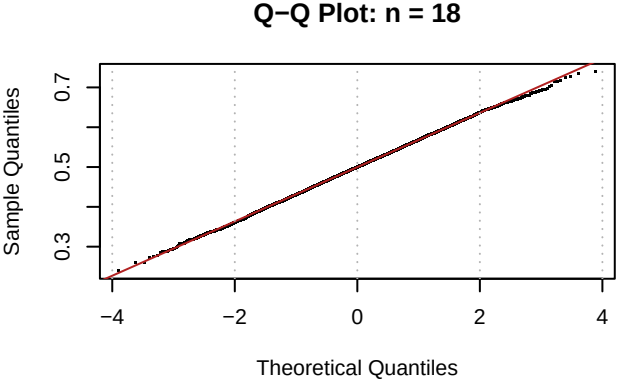

Sampling Distribution of Mean: n = 16
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

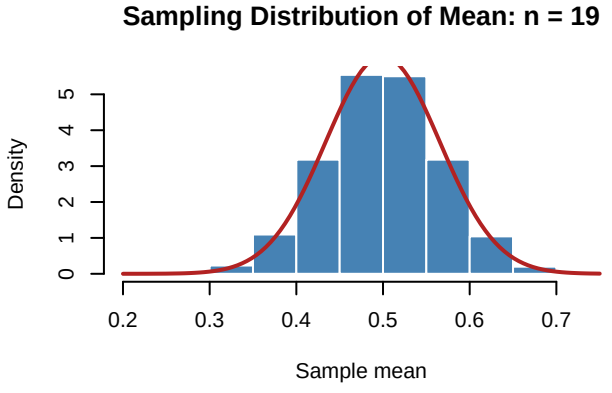

Q−Q Plot: n = 16
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

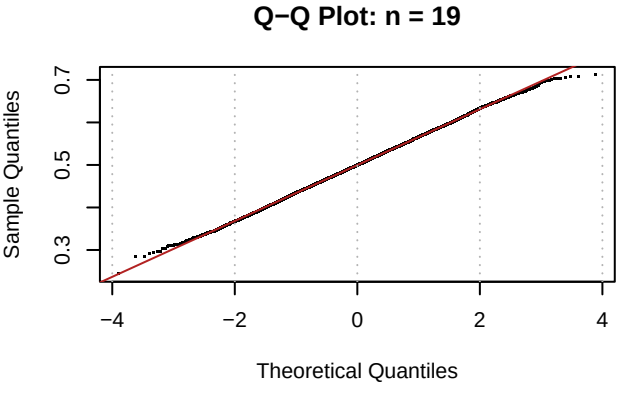

Sampling Distribution of Mean: n = 17
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

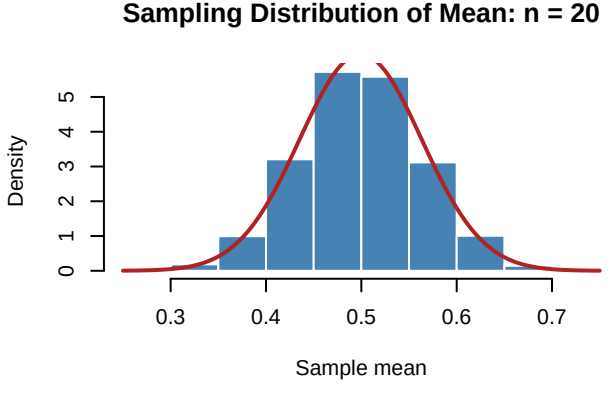

Q−Q Plot: n = 17
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

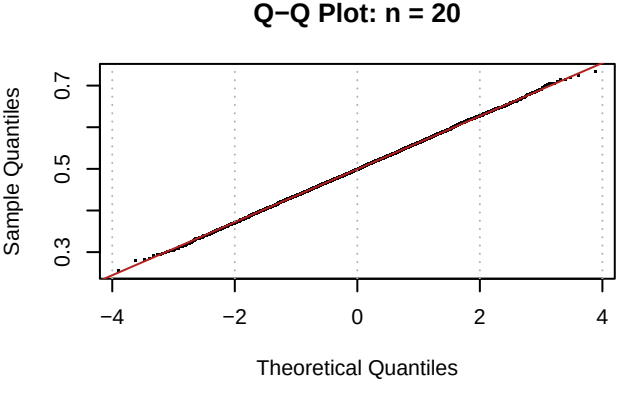

Sampling Distribution of Mean: n = 18
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
Sample mean

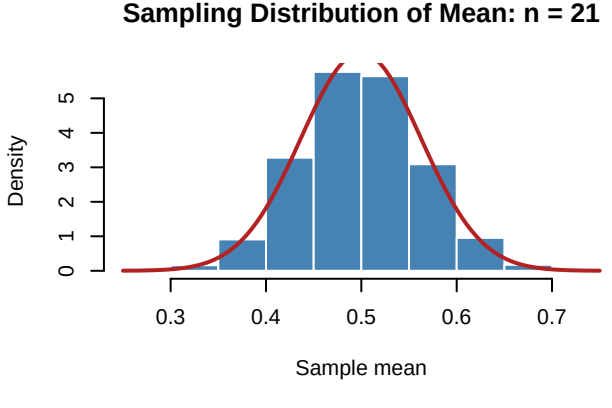

Q−Q Plot: n = 18
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

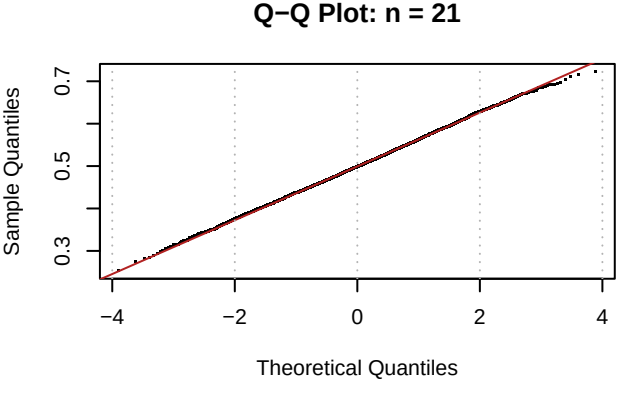

Sampling Distribution of Mean: n = 19
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
Sample mean

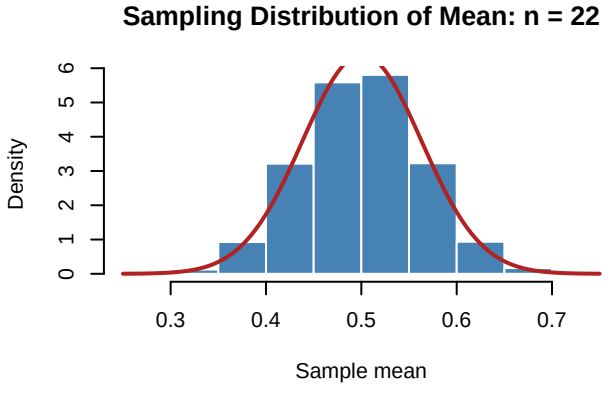

Q−Q Plot: n = 19
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

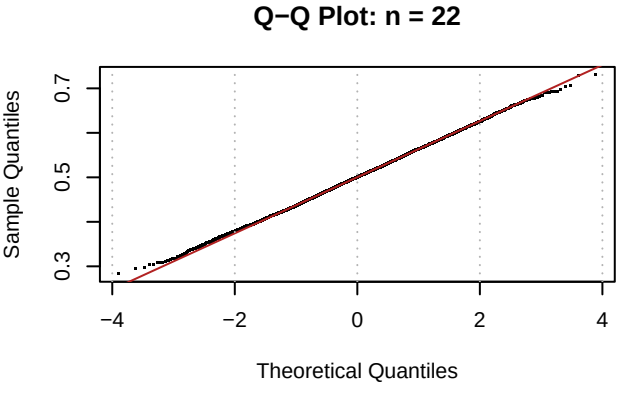

Sampling Distribution of Mean: n = 20
Density
0
1
2
3
4
5
0.3
0.4
0.5
0.6
0.7
Sample mean

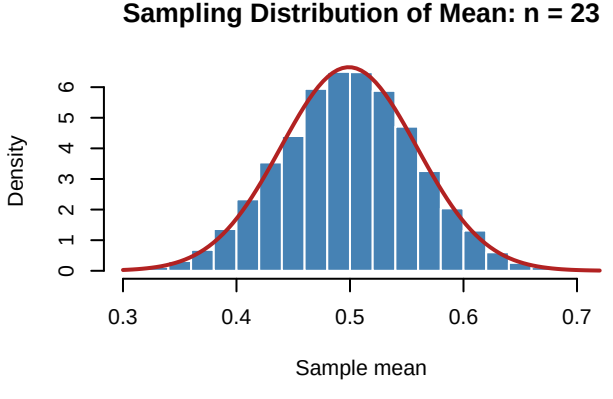

Q−Q Plot: n = 20
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

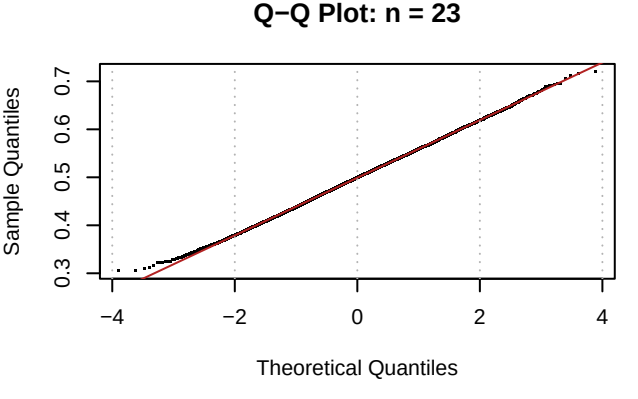

Sampling Distribution of Mean: n = 21
Density
0
1
2
3
4
5
0.3
0.4
0.5
0.6
0.7
Sample mean

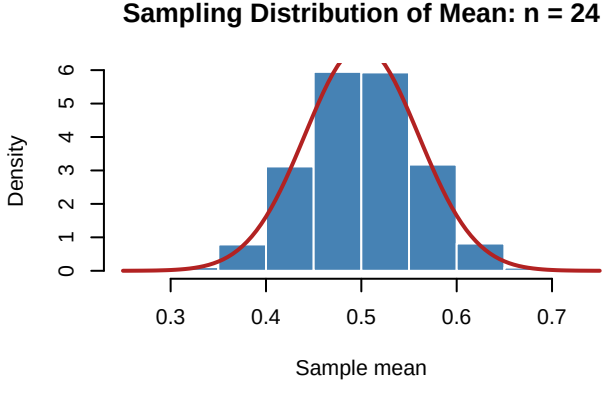

Q−Q Plot: n = 21
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

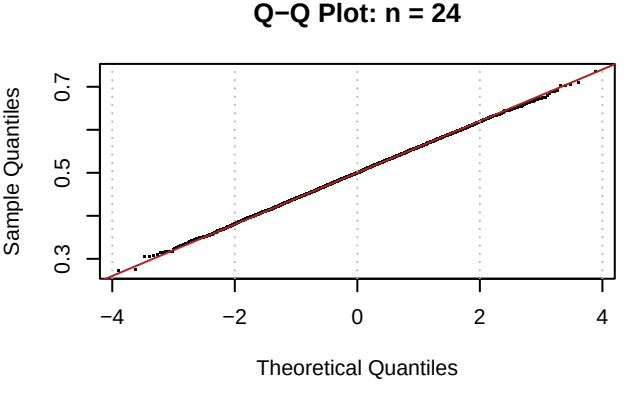

Sampling Distribution of Mean: n = 22
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

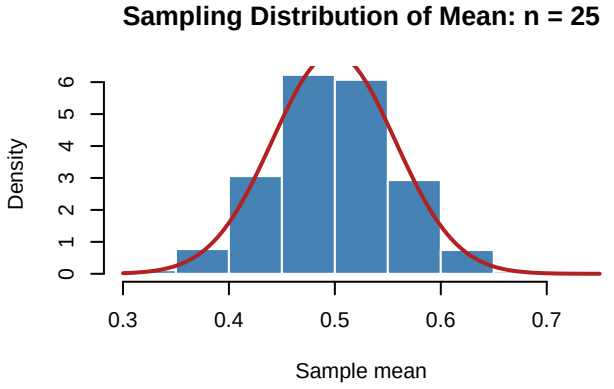

Q−Q Plot: n = 22
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

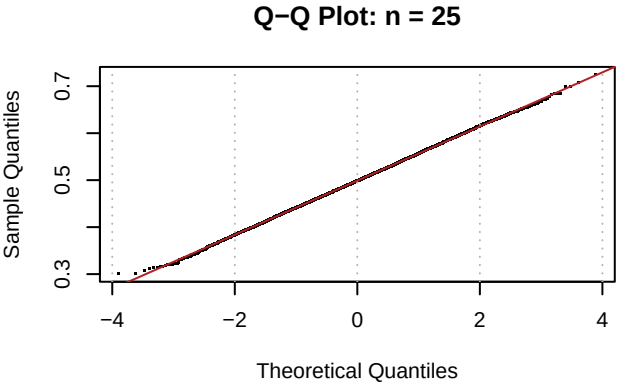

Sampling Distribution of Mean: n = 23
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

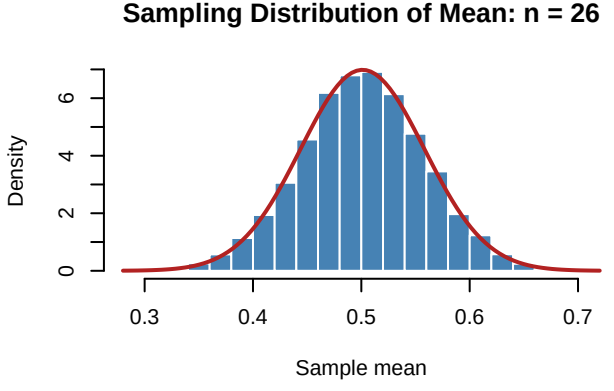

Q−Q Plot: n = 23
Sample Quantiles
0.3 0.4 0.5 0.6 0.7
−4 −2 0 2 4
Theoretical Quantiles

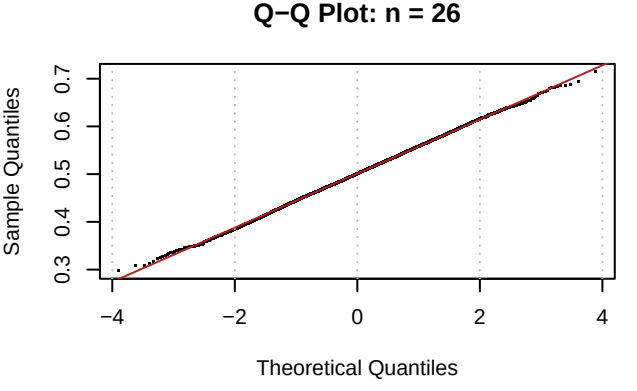

Sampling Distribution of Mean: n = 24
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

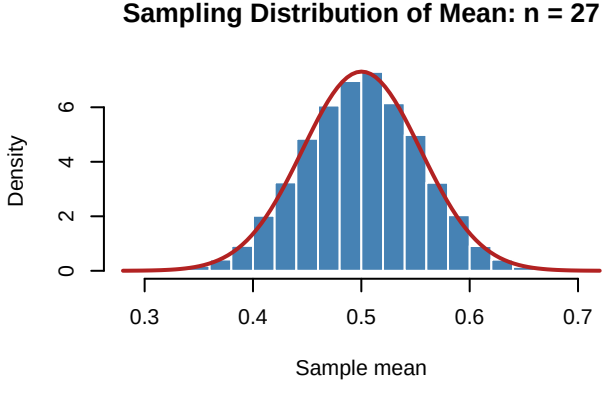

Q−Q Plot: n = 24
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

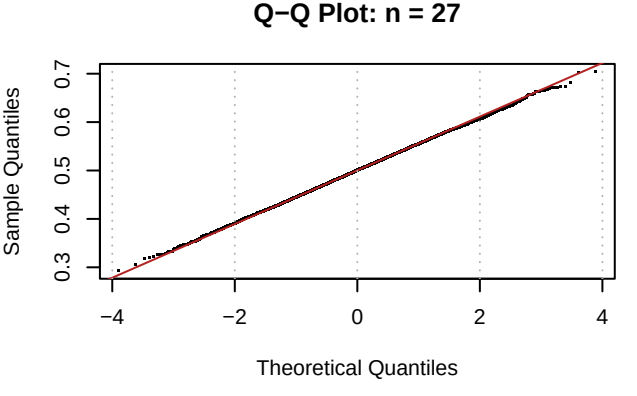

Sampling Distribution of Mean: n = 25
Density
0
1
2
3
4
5
6
0.3
0.4
0.5
0.6
0.7
Sample mean

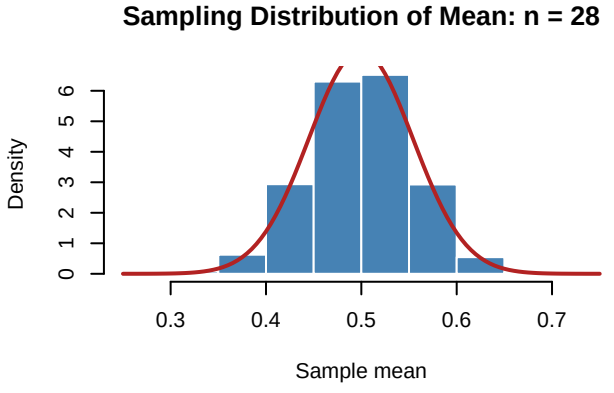

Q−Q Plot: n = 25
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

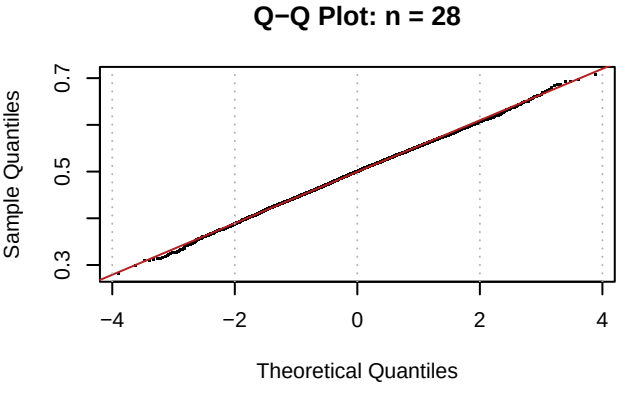

Sampling Distribution of Mean: n = 26
Density
0
2
4
6
0.3
0.4
0.5
0.6
0.7
Sample mean

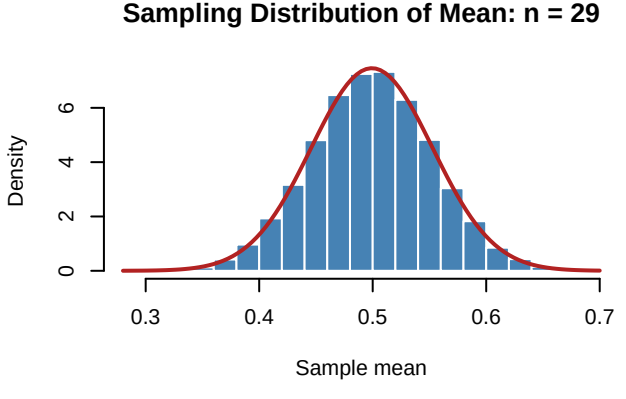

Q−Q Plot: n = 26
Sample Quantiles
0.3
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

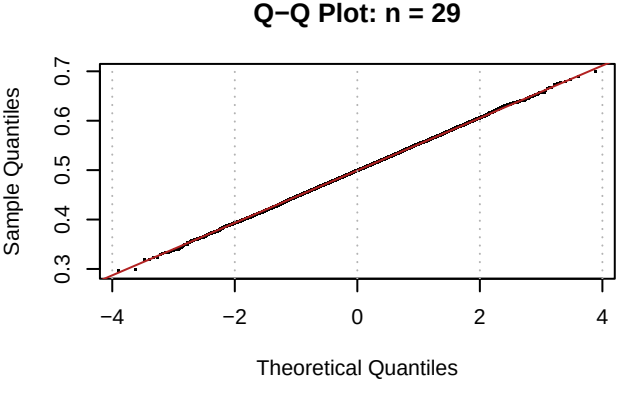

Sampling Distribution of Mean: n = 27
Density
0
2
4
6
0.3
0.4
0.5
0.6
0.7
Sample mean

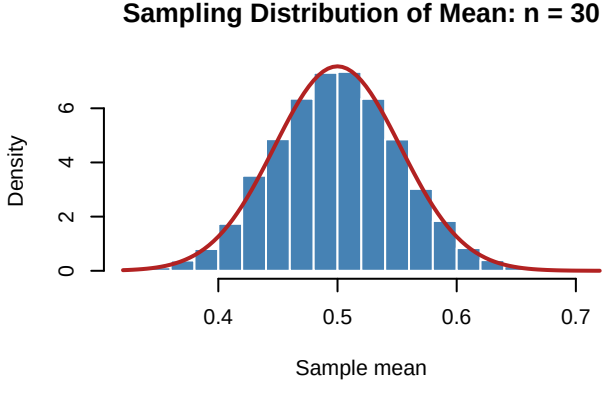

Q−Q Plot: n = 27
Sample Quantiles
0.3
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

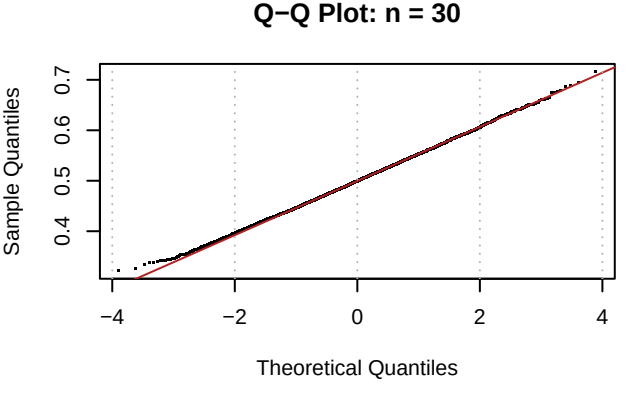

Sampling Distribution of Mean: n = 28
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

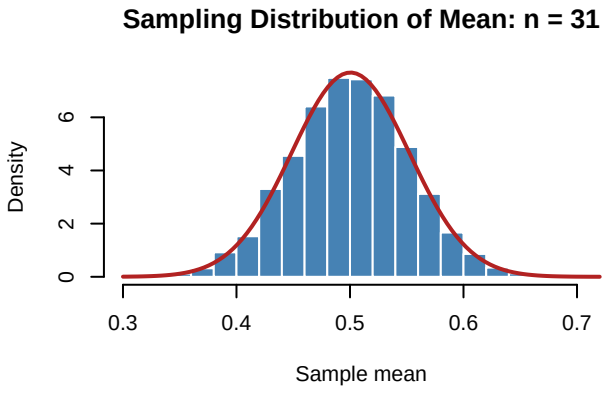

Q−Q Plot: n = 28
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

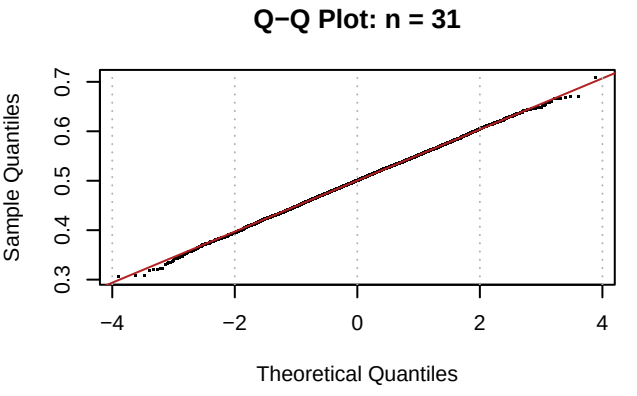

Sampling Distribution of Mean: n = 29
Density
0 2 4 6
0.3 0.4 0.5 0.6 0.7
Sample mean

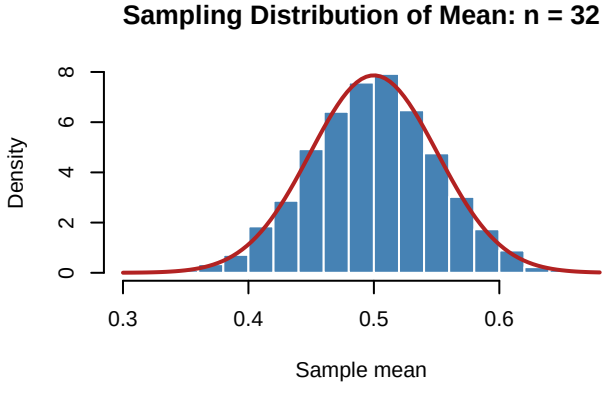

Q−Q Plot: n = 29
Sample Quantiles
0.3 0.4 0.5 0.6 0.7
−4 −2 0 2 4
Theoretical Quantiles

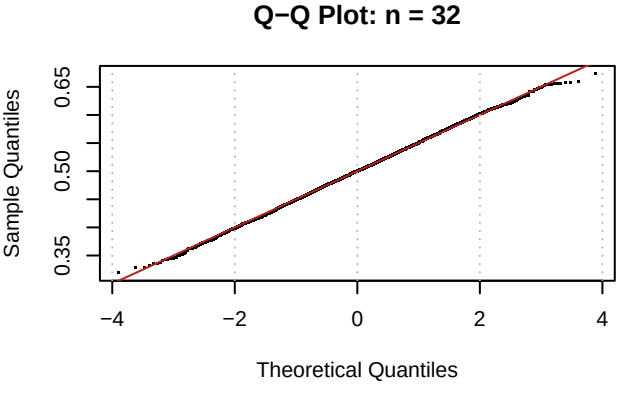

Sampling Distribution of Mean: n = 30
Density
0 2 4 6
0.4 0.5 0.6 0.7
Sample mean

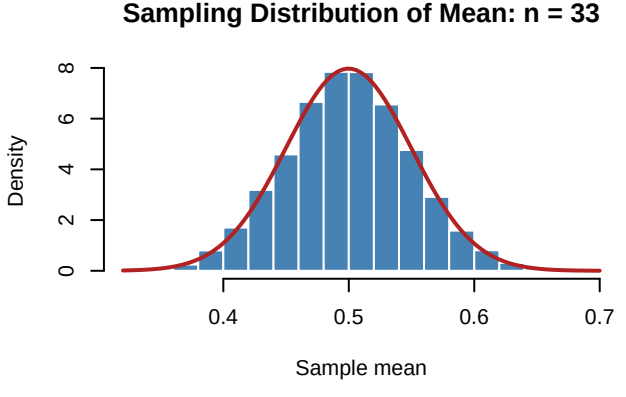

Q−Q Plot: n = 30
Sample Quantiles
0.4 0.5 0.6 0.7
−4 −2 0 2 4
Theoretical Quantiles

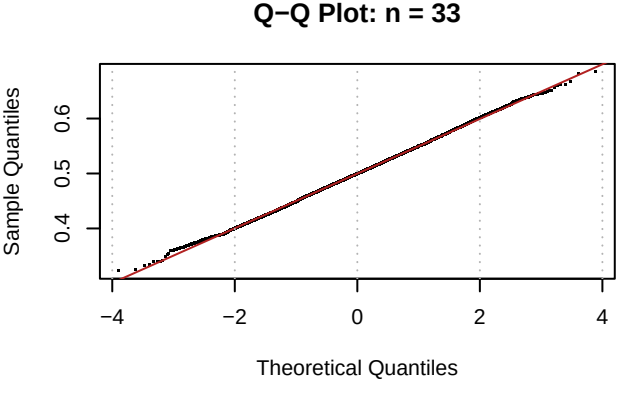

Sampling Distribution of Mean: n = 31
Density
0
2
4
6
0.3
0.4
0.5
0.6
0.7
Sample mean

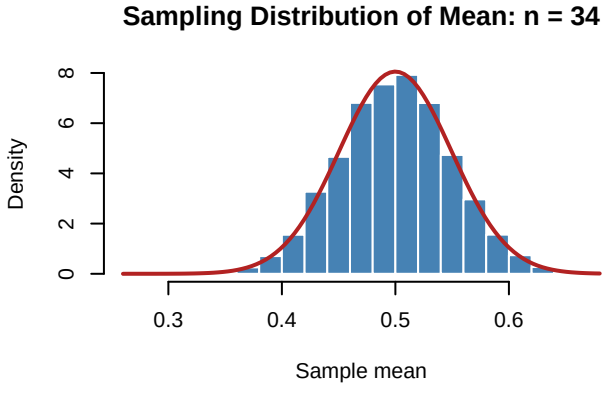

Q−Q Plot: n = 31
Sample Quantiles
0.3
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

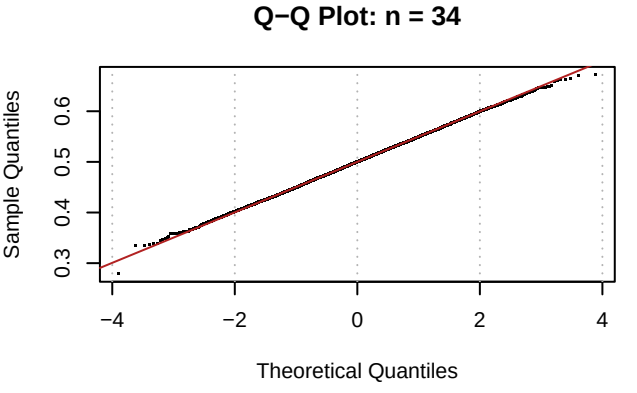

Sampling Distribution of Mean: n = 32
Density
0
2
4
6
8
0.3
0.4
0.5
0.6
Sample mean

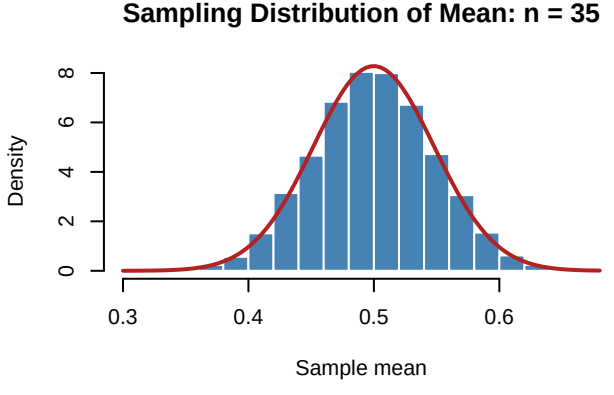

Q−Q Plot: n = 32
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

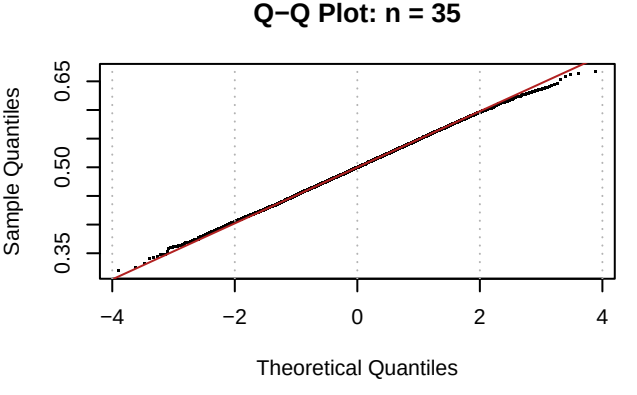

Sampling Distribution of Mean: n = 33
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

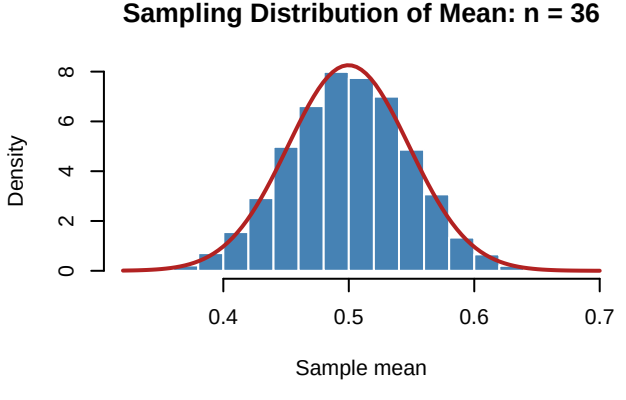

Q−Q Plot: n = 33
Sample Quantiles
0.4
0.5
0.6
−4
−2
0
2
4
Theoretical Quantiles

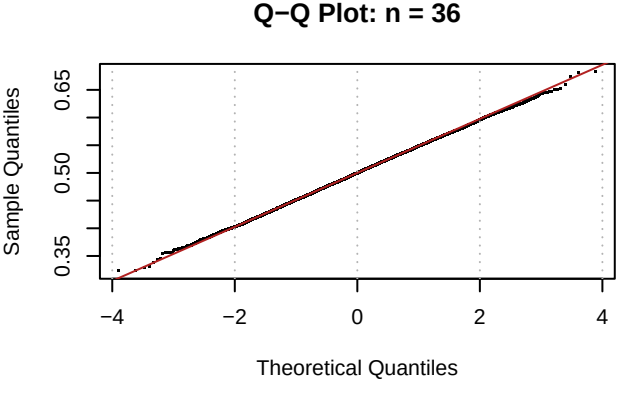

Sampling Distribution of Mean: n = 34
Density
0
2
4
6
8
0.3
0.4
0.5
0.6
Sample mean

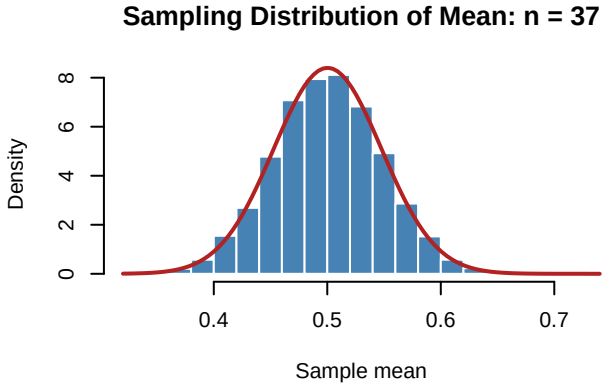

Q−Q Plot: n = 34
Sample Quantiles
0.3
0.4
0.5
0.6
−4
−2
0
2
4
Theoretical Quantiles

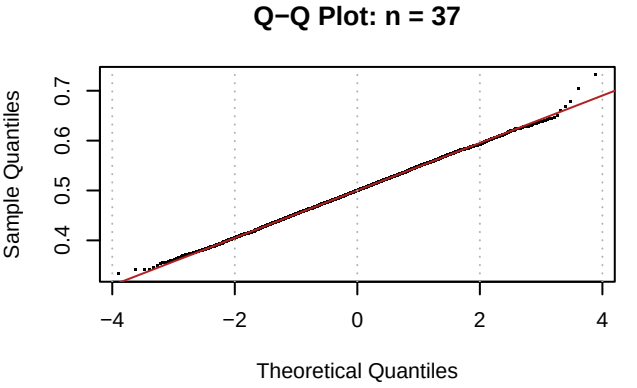

Sampling Distribution of Mean: n = 35
Density
0
2
4
6
8
0.3
0.4
0.5
0.6
Sample mean

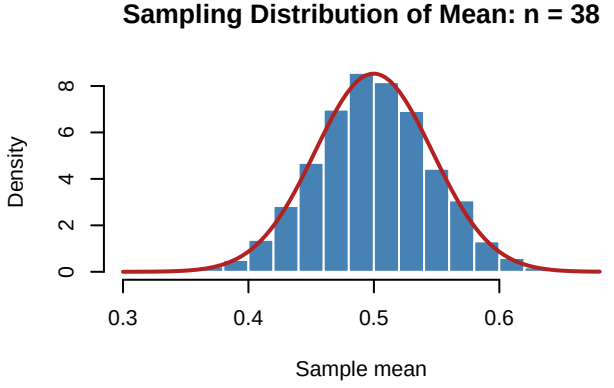

Q−Q Plot: n = 35
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

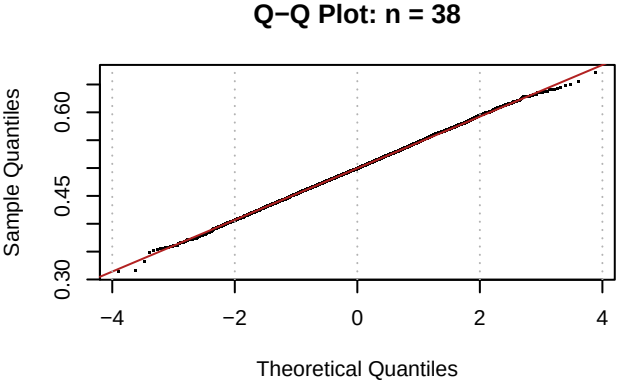

Sampling Distribution of Mean: n = 36
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

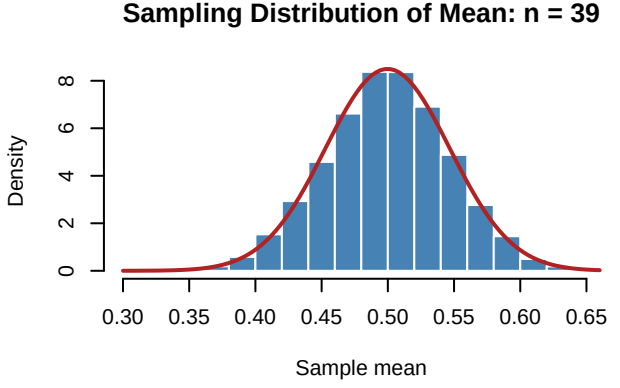

Q−Q Plot: n = 36
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

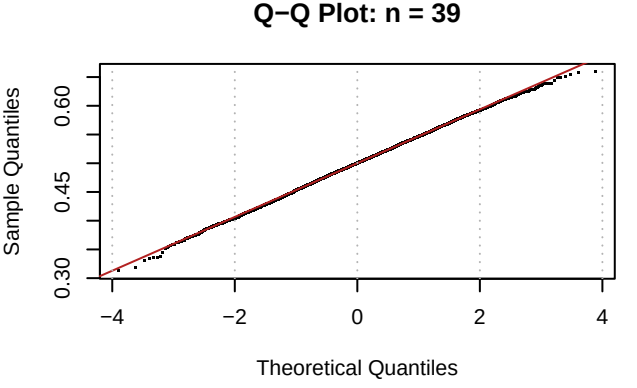

Sampling Distribution of Mean: n = 37
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

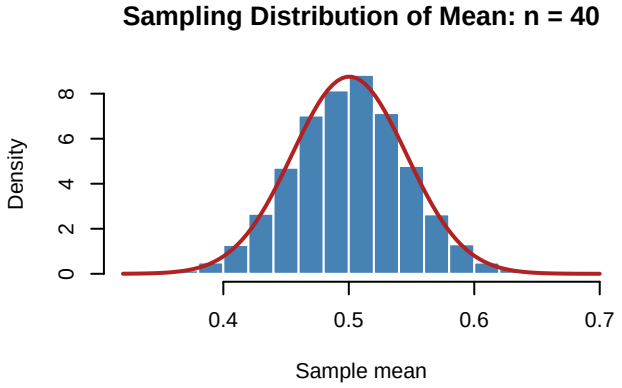

Q−Q Plot: n = 37
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

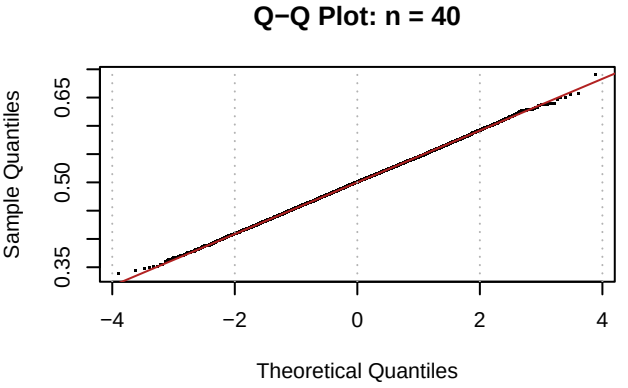

Sampling Distribution of Mean: n = 38
Density
0
2
4
6
8
0.3
0.4
0.5
0.6
Sample mean

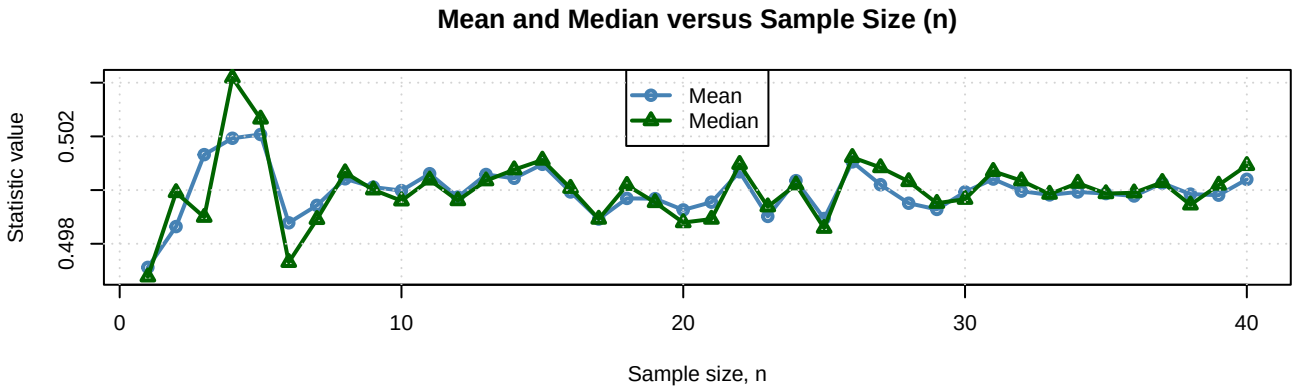

Q−Q Plot: n = 38
Sample Quantiles
0.30
0.45
0.60
−4
−2
0
2
4
Theoretical Quantiles

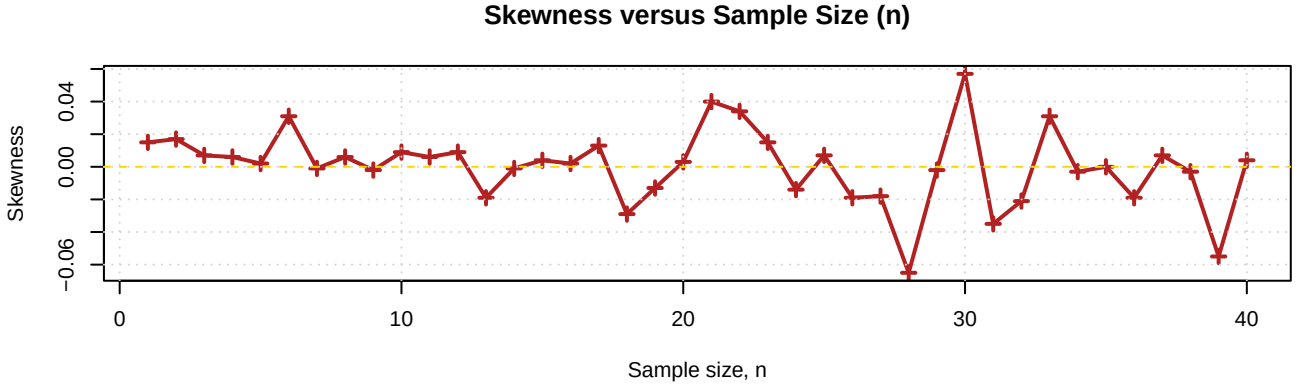

Sampling Distribution of Mean: n = 39
Density
0
2
4
6
8
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

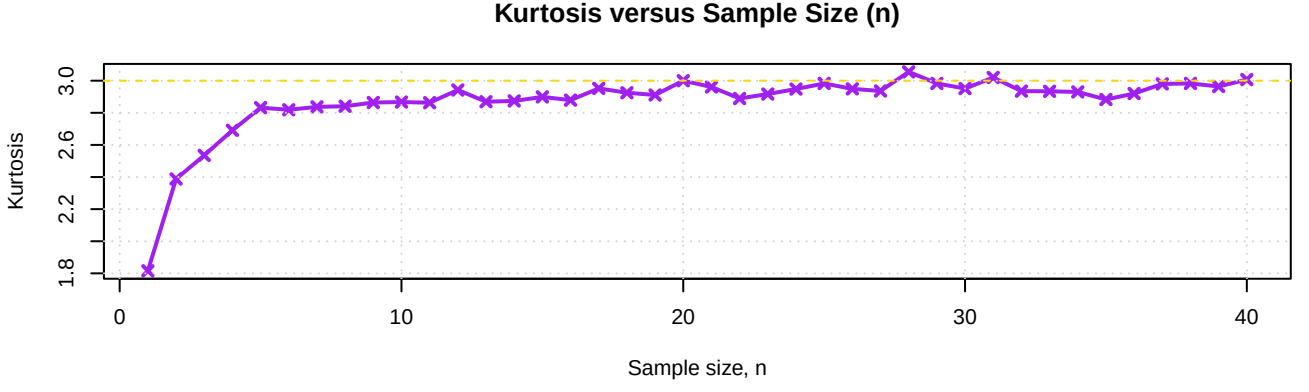

Q−Q Plot: n = 39
Sample Quantiles
0.30
0.45
0.60
−4
−2
0
2
4
Theoretical Quantiles


Sampling Distribution of Mean: n = 40
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean


Q−Q Plot: n = 40
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles


Mean and Median versus Sample Size (n)
Mean
Median
Statistic value
0.498
0.502
0
10
20
30
40
Sample size, n


Skewness versus Sample Size (n)
Skewness
−0.06
0.00
0.04
0
10
20
30
40
Sample size, n


Kurtosis versus Sample Size (n)
Kurtosis
1.8
2.2
2.6
3.0
0
10
20
30
40
Sample size, n